NeuroSynIQ | Testing & Validation Evidence

Science Fair 2026 | Innovation Category | Research & Development Prototype

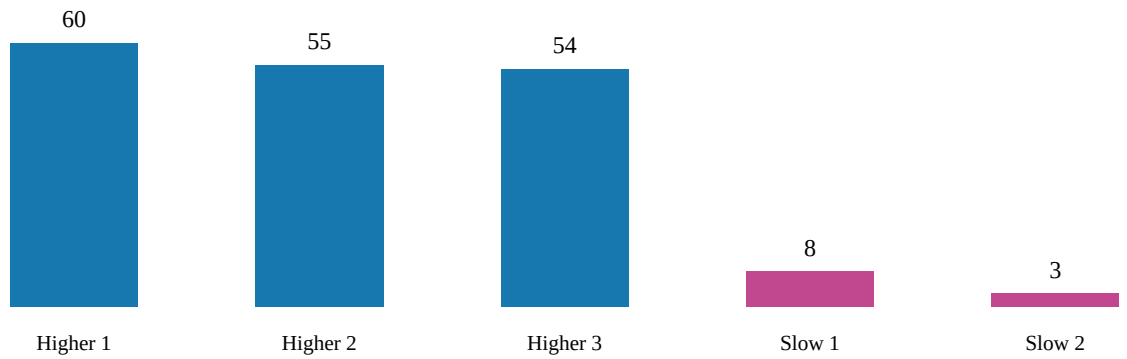
1. Testing objective

Evaluate whether the prototype measures an intentionally introduced tapping change, and whether its PersistenceGuard rules distinguish temporary simulated changes from repeated or persistent simulated changes.

2. Real tablet experiment — 9 October 2026

Method: One participant, right dominant hand, five 10-second Tap Tests. Three higher-count trials were compared with two deliberately slower trials. This was a small, non-randomized, retrospective comparison.

Condition	Trial 1	Trial 2	Trial 3	Average
Higher-count	60	55	54	56.3
Deliberately slower	8	3	—	5.5



Observed: All three higher-count trials were within baseline; both deliberately slower trials were flagged as measured changes. Average tapping fell by approximately 90.2% (56.3 to 5.5 taps).

Interpretation: This shows responsiveness to intentional slowing in this sample. It does not establish clinical accuracy or disease detection.

3. 100-case software simulation

Purpose: Compare a single-session threshold with PersistenceGuard, which distinguishes an observed change (one session), a repeated change (two consecutive sessions), and persistent change (three or more consecutive sessions).

Design: 100 simulated cases across five conditions, 20 each: stable, normal variation, one-off temporary change, gradual change, and persistent change. Each case was evaluated over five sessions.

Simulated outcome	Simple threshold	PersistenceGuard
Gradual/persistent patterns flagged as repeated or persistent	40/40 (single-session flag)	40/40
Strict persistence within five sessions	Not equivalent	37/40
Temporary one-off cases treated as persistent	20/20	0/20
Gradual/persistent cases without strict confirmation by session 5	0/40 (immediate flag)	3/40

Finding: PersistenceGuard did not strictly label any of the 20 one-off simulated cases as persistent. It strictly confirmed 37/40 gradual or persistent cases within five sessions; three needed further observation.

Trade-off: Requiring consecutive changes reduces premature persistence labels but delays strict confirmation of some gradual changes.

4. Limitations

• Real experiment: one participant, only five trials, unequal trial counts (3 vs 2), no randomization, and no independently documented device, finger, or rest controls.
• Deliberate slowing does not represent a neurological disease.
• Other historical measurements mix participants and are not a same-person control group.
• The 100 cases are simulations, not patient tests; the figures do not represent clinical sensitivity or specificity.
• PersistenceGuard is rule-based monitoring logic, not a clinically trained diagnostic model.

5. Engineering conclusion

The tablet trial provides preliminary evidence of measurement and change flagging. The simulation demonstrates how persistence logic behaves across constructed scenarios. NeuroSynIQ remains a research prototype intended to support review of changes, not to diagnose disease or replace a clinician.

6. Suggested judges' demonstration

Show Tap Test → MotorPrint baseline → a measured-change comparison → the WHY? explanation → Research Lab scenarios. Clearly label simulated demonstration data.

Evidence source: NeuroSynIQ “Testing & Empirical Data” trifold record, including the 9 October 2026 experiment and 100-case validation summary. No new experiments were conducted to prepare this PDF.

RESEARCH PROTOTYPE | Does not diagnose disease or replace a clinician.